

for further analysis. Once intercepted RF signal is monitored and recorded, the system investigates RF spectrum to identify the target message content.

Who needs it

Surveillance of wireless signals is generally required by spectrum regulators, public safety agencies, border and coastal security, and military intelligence. In the government and military areas, these transmissions are often characterised as signals intelligence (SIGINT). These systems are useful to intercept and locate the different wireless transmitters such as global mobile personal communications by satellite (GMPCS), cellular handsets and VHF/UHF frequency range walkie-talkies. Thus, an RF surveillance system plays a vital role and ensures success in security mission.

Various intelligence and security department personnel like Homeland security department, Border and Coastal Surveillance department, critical infrastructure and perimeter protection forces, signal intelligence groups of Army, Navy and Air Force, technical and aviation research organisations, defence agencies, government end-users, paramilitary forces, regulatory agencies, commercial spectrum monitoring agencies, railways, hos-

pitals, transportation security department, including aviation and maritime transportation, can use these systems.

Overview of use case

Let us define various use cases for RF surveillance systems in different operating environments. Three use cases are stated below:

1. Tactical use case
2. Preventive use case
3. Costal surveillance use case

Note that these are just indicative use patterns, and the RF surveillance system is not limited to these applications.

Tactical use case for critical infrastructure protection

RF sensors can be installed in the suspected area around the building as shown in Fig. 2, or RF sensors can be mounted in the sensitive buildings as shown in Fig. 3. All RF sensors can be connected via LAN backhaul either with a wired or wireless system. The sensors can then be time-synchronised and, with its core detection algorithm, the software can detect illegal or threat signal. The system is very easy to install within a few minutes.

This method is best explained with the example of Hotel Taj Mahal in

Mumbai where terrorists attacked. Assuming that the terrorists were still using RF communication for their coordination, it was possible to detect their signals and their precise origination, if multiple RF sensors (Fig. 2) were put in all corners of hotel building.

Using cross-correlation technique the exact RF origination or emitting source, and hence terrorists' locations, could be detected and they could be isolated and neutralised.

Though this technique is reactive and tactical, it can be really quick to spot the emitter and corrective actions can be taken to avoid larger damage. A better way, however, is to proactively monitor indoor areas of the sensitive buildings.

Fig. 3 shows a large hotel building with multiple floors and rooms. Each RF sensor scans its designated area and sends information to a common control location where all the RF-emitting sources (legal or illegal) in the hotel can be identified. Since sensors can time-synchronise their sweeps, the energy detection too gets synchronised. Power-level comparisons can then determine whether a transmitter is inside vs outside, and if inside, in which room.

This approach works well even if attenuation through the walls, ceilings and floors of buildings is substantial. The series of actions performed by this system in current example is summarised in Fig. 4.

Preventive use case for critical infrastructure protection

In preventive application of an infrastructure protection, a distributed network of RF surveillance systems can be used to perform the operation in remote mode, probably from network operations centre (NOC). The surveillance systems can be installed at various locations by connecting them on backhaul connectivity such as the Internet, 3G or a captive LAN network. These systems stay synchronised in time with GPS and therefore have the capability to geo-map suspicious emitters.

Let us take the example of Mumbai

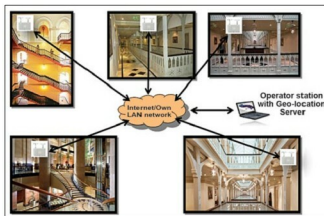


Fig. 3: Use case of proactive indoor monitoring of sensitive areas



Fig. 4: Activity sequence for commissioning of RF surveillance system